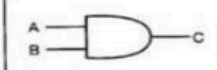
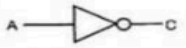
EXPERIMENT NO.1

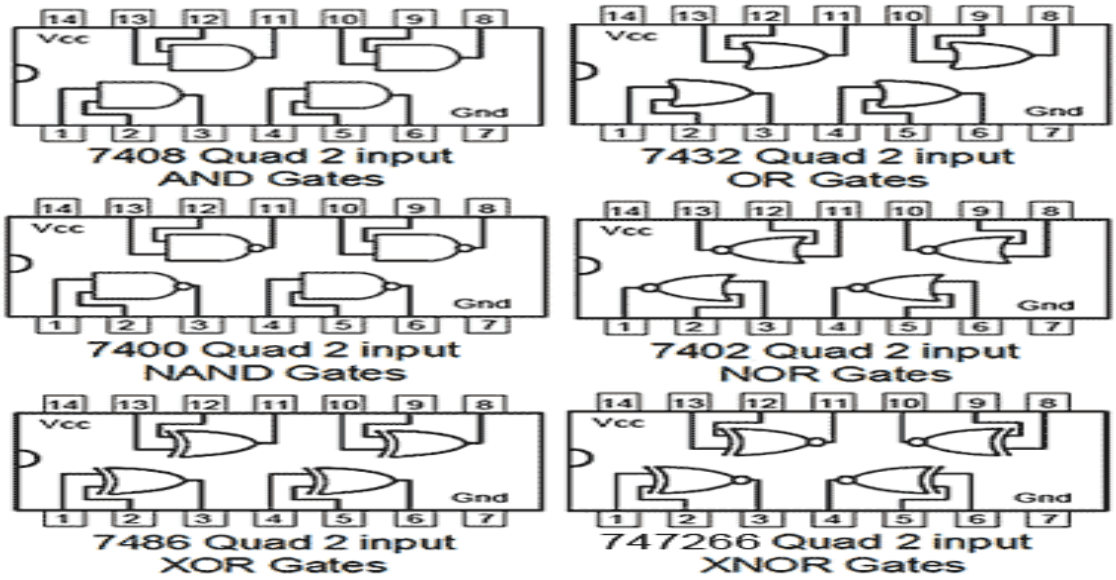
1a) Aim : Study and verify the truth table of logic gates.

Components Required: Trainer Kit, Patch Cords and ICs.

SL.NO	IC
1	7408
2	7432
3	7404
4	7400
5	7402
6	7486

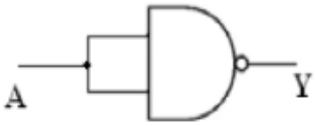
Name	Description	Logic gate	Truth table
AND	The output value of C is 1 when both input values A and B are 1.		Input A: 0 0 1 1; Input B: 0 1 0 1; Output C = AB: 0 0 0 1
OR	If either of the input values are 1 then the output will be 1.		Input A: 0 0 1 1; Input B: 0 1 0 1; Output C = A+B: 0 1 1 1
NOT	The state of the output is the inverse of the input.		Input A: 0 1; Output C = $\bar{A}$: 1 0
NAND	Not AND. The output value C is 1 except when both A and B are 1.		Input A: 0 0 1 1; Input B: 0 1 0 1; Output C = $(\overline{AB})$: 1 1 1 0
NOR	The output C is 1 when both inputs A and B are 0.		Input A: 0 0 1 1; Input B: 0 1 0 1; Output C = $(\overline{A+B})$: 1 0 0 0
XOR	The output C is 1 except when both inputs A and B are the same.		Input A: 0 0 1 1; Input B: 0 1 0 1; Output C = $A \oplus B$: 0 1 1 0

Pinout :

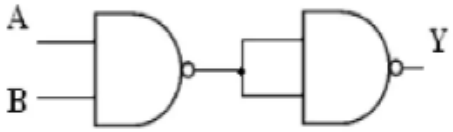


1b) Realization using Universal Gates

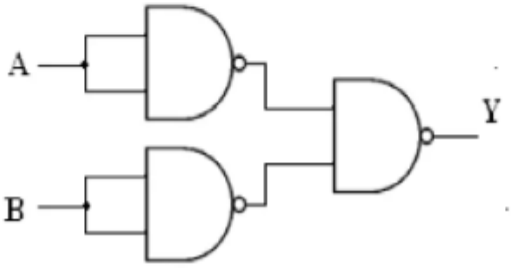
(a) NOT gate: $Y = A'$



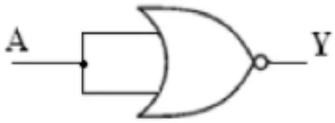
(b) AND gate: $Y = A \cdot B$



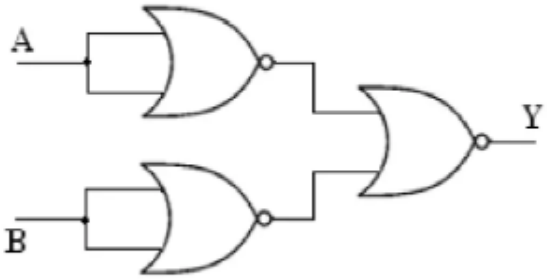
(c) OR gate: $Y = A + B$



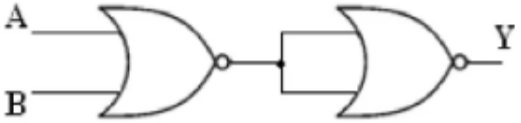
(a) NOT gate: $Y = A'$



(b) AND gate: $Y = A \cdot B$

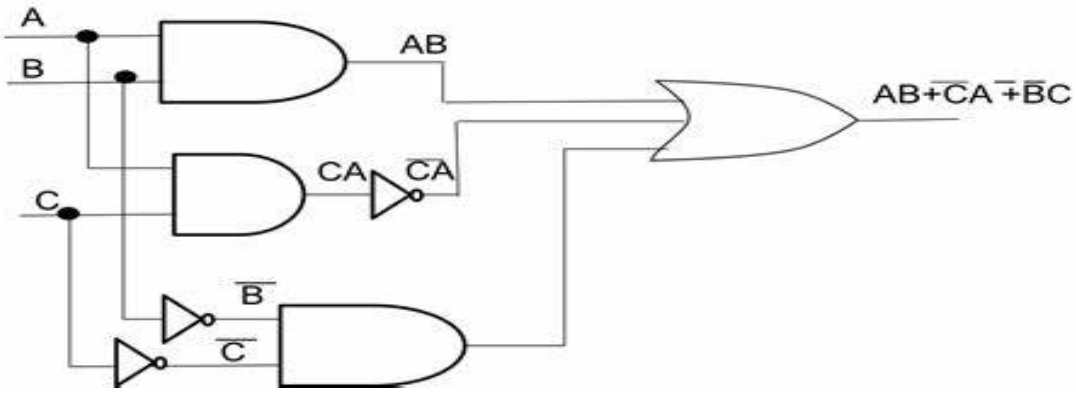


(c) OR gate: $Y = A + B$



1c) Simplify the given expression and realize it using gates.

$AB+\overline{CA}+\overline{B}C$



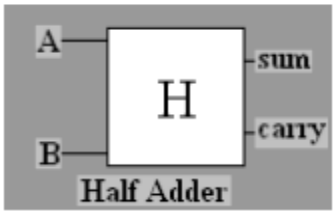
Results: The logic gates are constructed and truth tables are verified.

EXPERIMENT NO.2

Aim : Design and realize
a) Half Adder / Full Adder.
b) Half Subtractor / Full Subtractor.

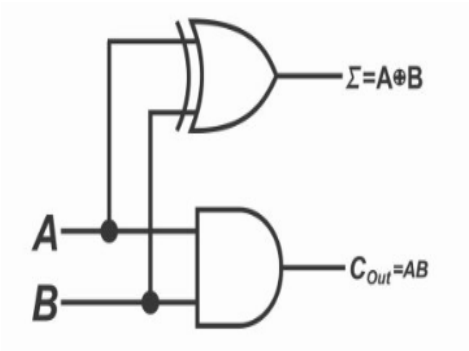
Half Adder

Logic Diagram

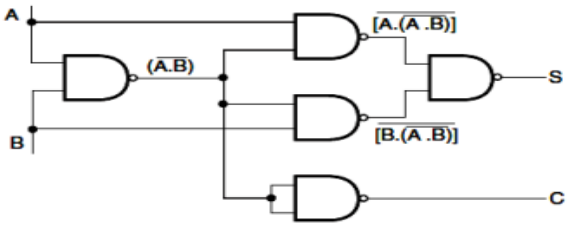


Truth Table

A	B	Sum (S)	Carry (C)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

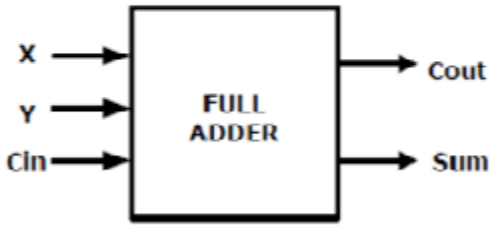


USING NAND GATES ONLY



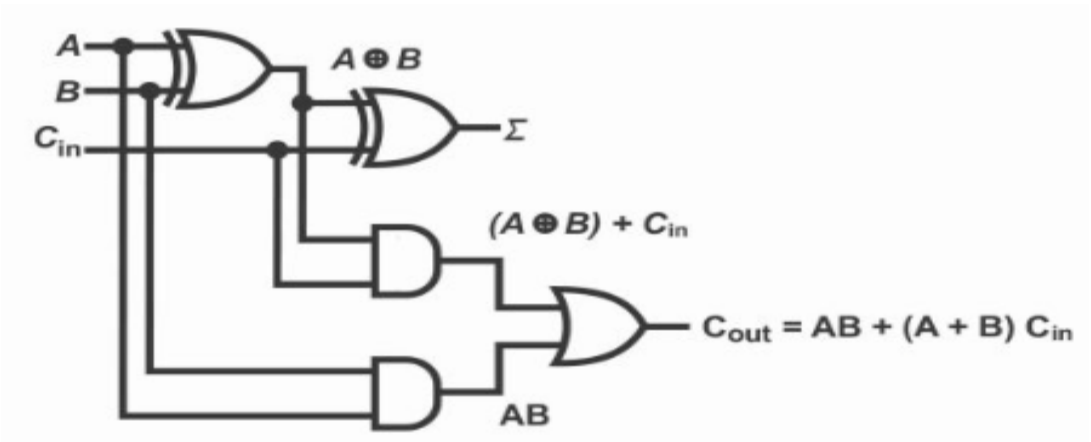
Full Adder

Logic Diagram

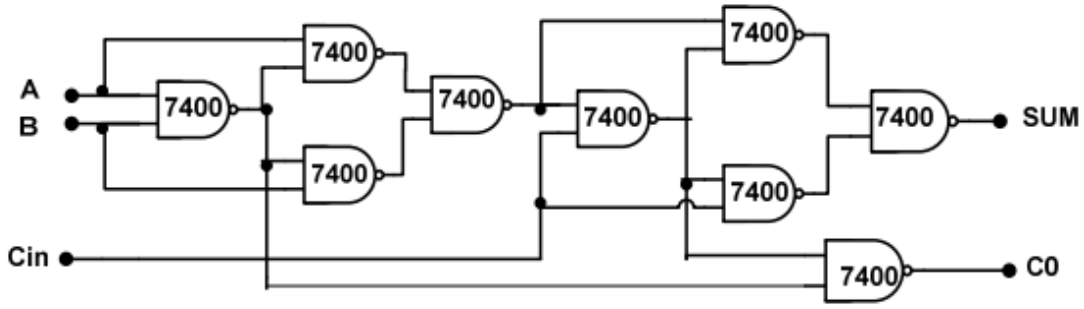


Truth Table

C_i	A	B	Sum (S)	Carry (C_o)
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

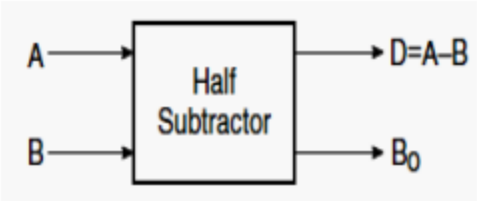


USING NAND GATES ONLY



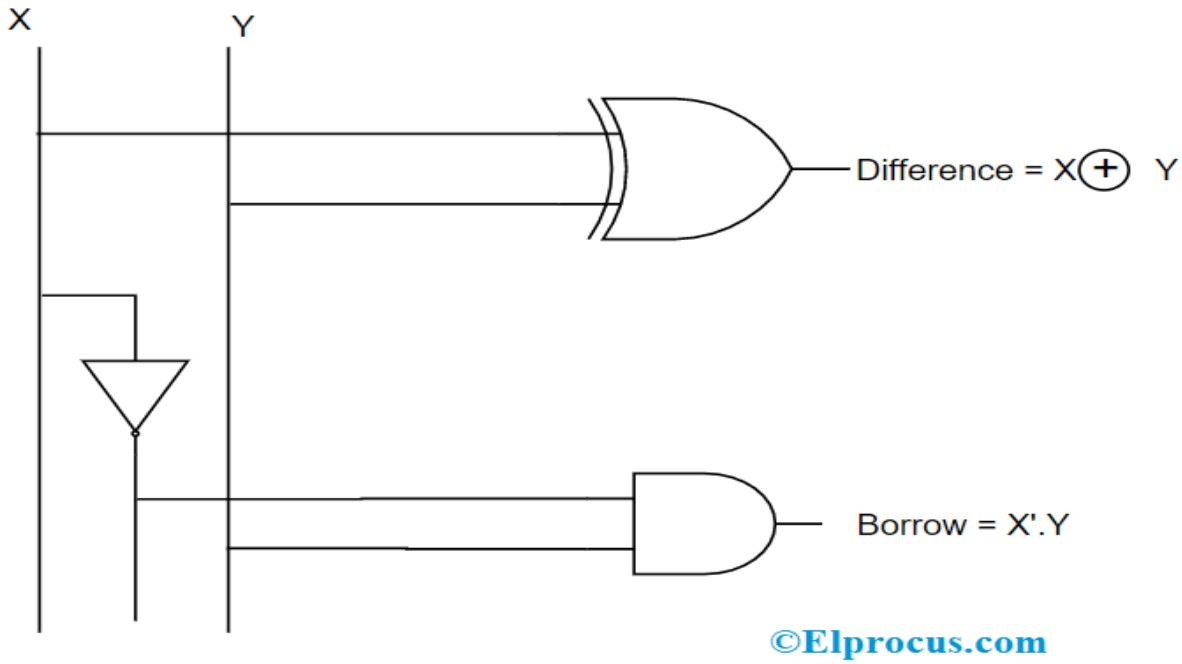
Half subtractor

Logic Diagram

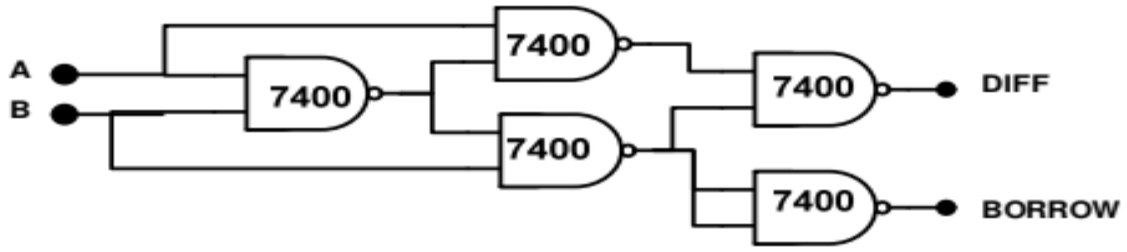


Truth Table

A	B	Diff (D)	Borrow (B_0)
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

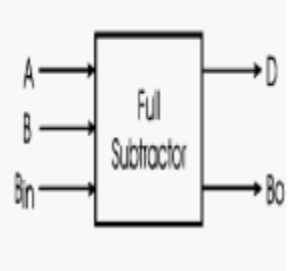


USING NAND GATES ONLY



Full subtractor

Logic Diagram:



Truth Table

Minuend (A)	Subtrahend (B)	Borrow In (B_{in})	Difference (D)	Borrow Out (B_O)
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

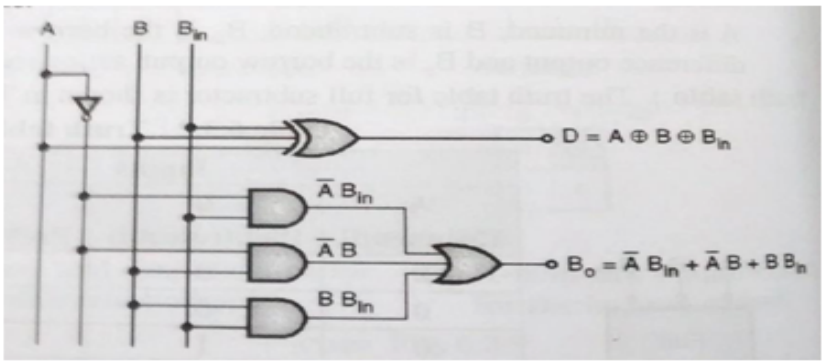
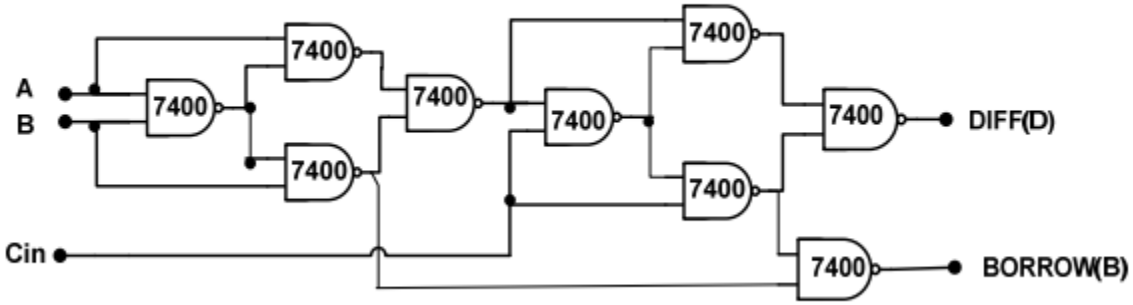


Fig. Logic diagram of a full subtractor

(USING NAND GATES ONLY)

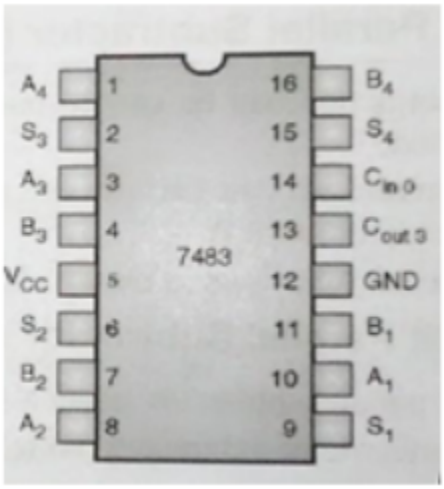


Results: The truth tables are verified.

EXPERIMENT NO.3

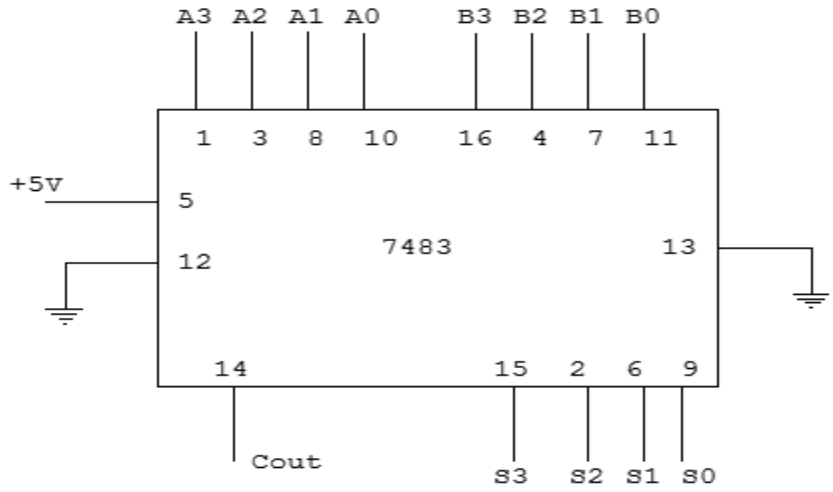
Design and set up the following circuit using IC 7483.
a. A 4-bit binary parallel adder.
b. A 4-bit binary parallel subtractor.

Pinout:

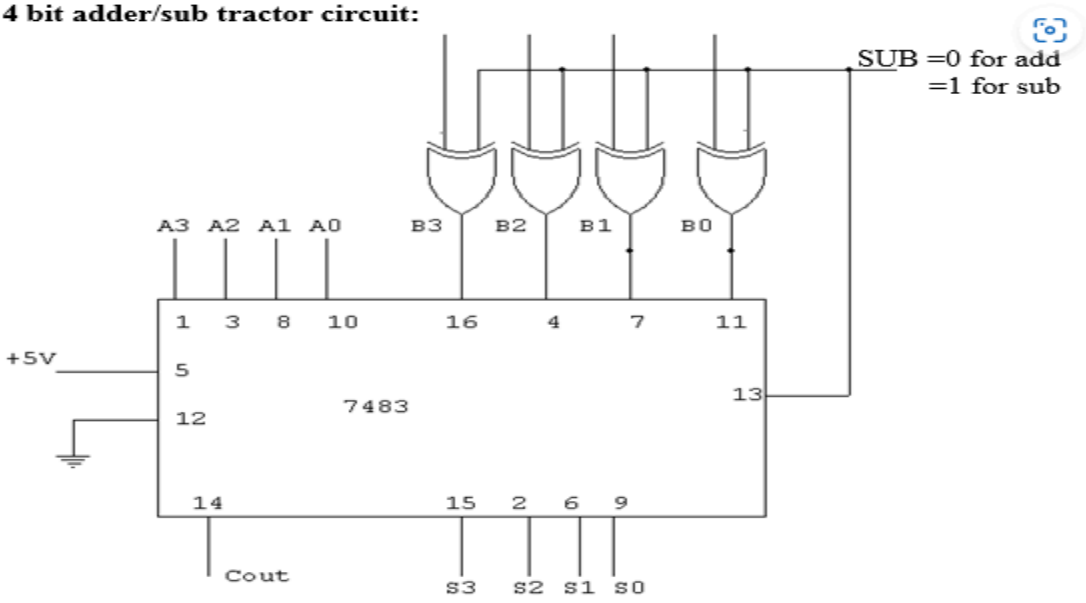


Circuit Diagrams:

4 bit adder circuit:



4 bit adder/sub tractor circuit:



Results: The logic gates are constructed and truth tables are verified.

EXPERIMENT NO.4

Design and realize the following using Basic gates & IC 7483.
a. BCD to Excess- 3 Code and vice versa
b. Binary to Gray Code and vice versa

Excess-3 to BCD:

Truth Table :

Inputs				Outputs			
W	X	Y	Z	A	B	C	D
0	0	1	1	0	0	0	0
0	1	0	0	0	0	0	1
0	1	0	1	0	0	1	0
0	1	1	0	0	0	1	1
0	1	1	1	0	1	0	0
1	0	0	0	0	1	0	1
1	0	0	1	0	1	1	0
1	0	1	0	0	1	1	1
1	0	1	1	1	0	0	0
1	1	0	0	1	0	0	1

MAP :

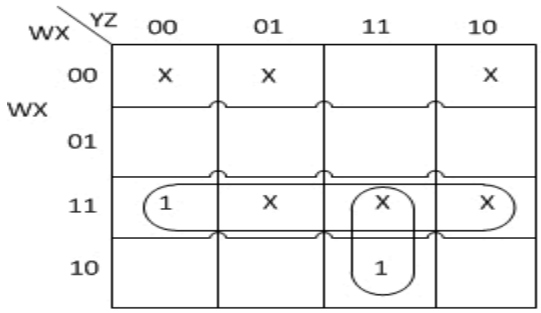


Fig: Map for A

$$A = WX + WYZ$$

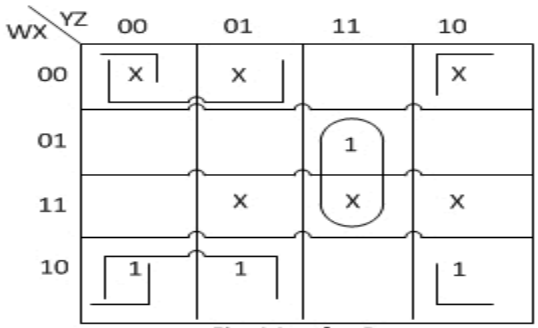


Fig: Map for B

$$B = XY' + XYZ + XZ'$$

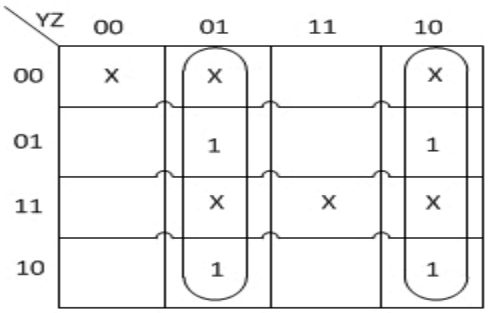


Fig: Map for C

$$C = Y'Z + YZ'$$

OR

$$C = Y \oplus Z$$

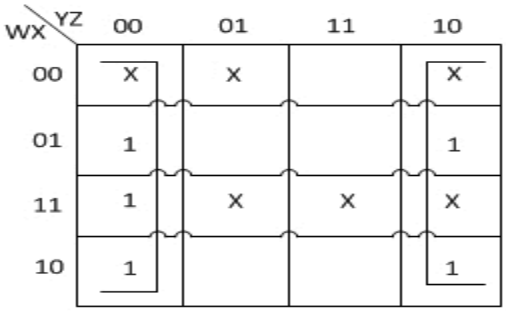
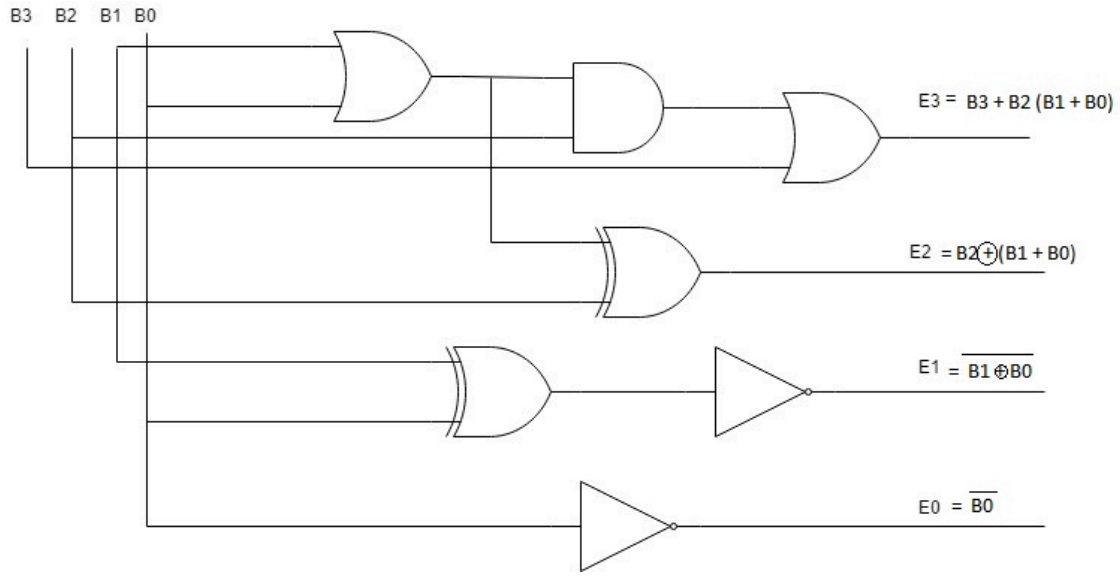
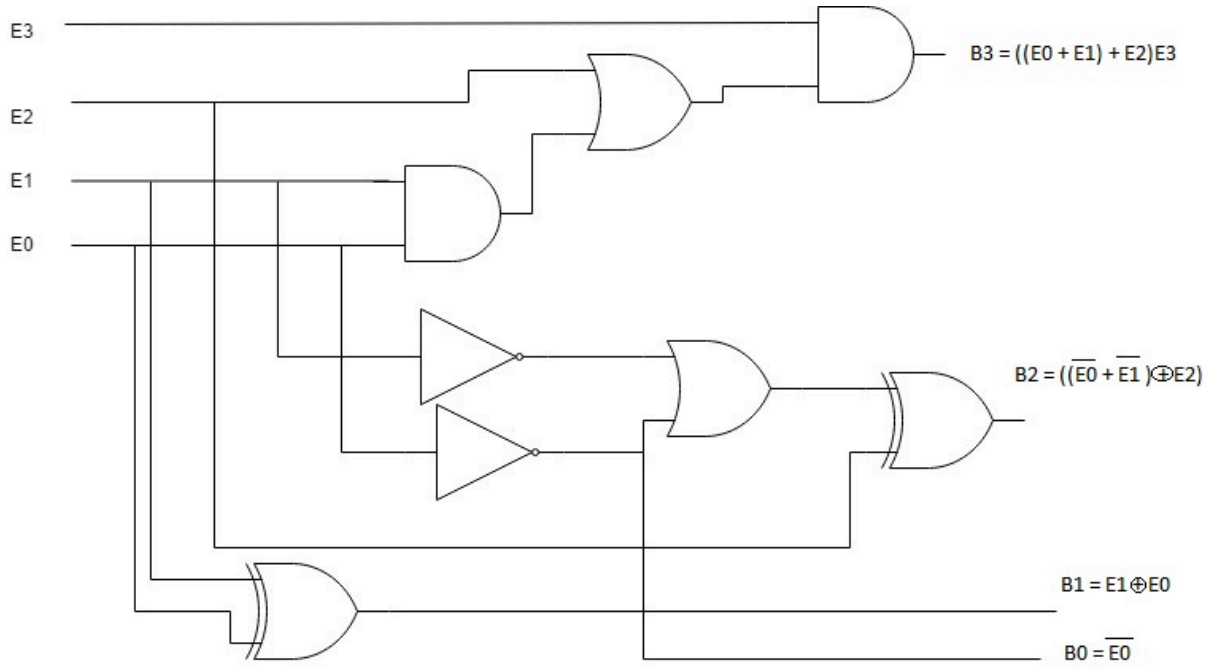
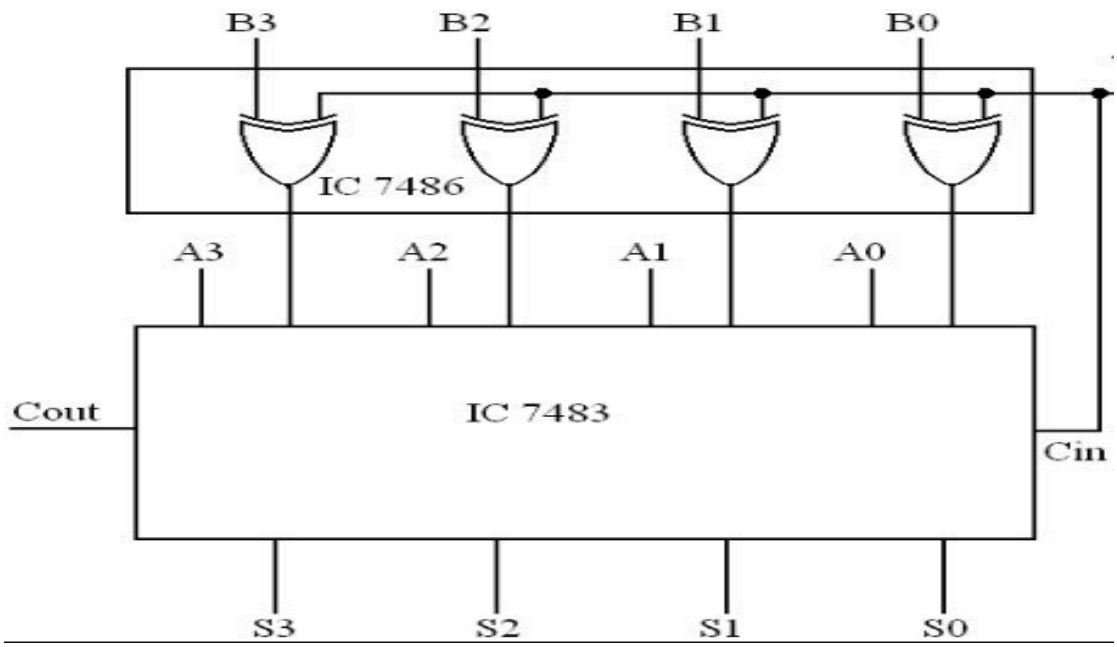


Fig: Map for D

$$D = Z'$$

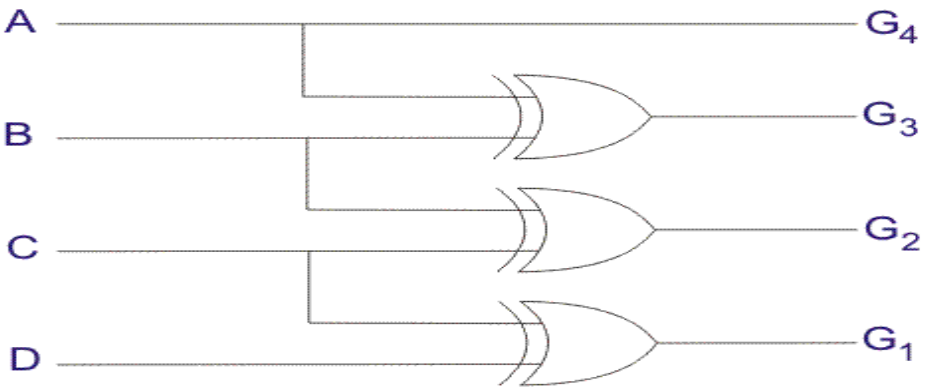

B3
B2
B1
B0
E3 = B3 + B2 (B1 + B0)
E2 = B2⊕(B1 + B0)
E1 = $\overline{B1 \oplus B0}$
E0 = $\overline{B0}$


E3
E2
E1
E0
B3 = ((E0 + E1) + E2)E3
B2 = (($\overline{E0}$ + $\overline{E1}$)⊕E2)
B1 = E1⊕E0
B0 = $\overline{E0}$

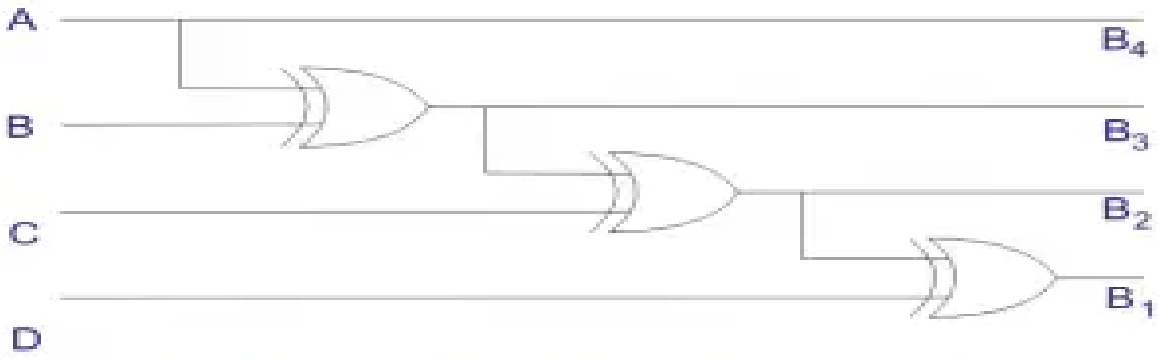


Binary to Gray Code:

BINARY INPUT				GRAY CODE OUTPUT			
B3	B2	B1	B0	G3	G2	G1	G0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	1
1	0	1	1	1	1	1	0
1	1	0	0	1	0	1	0
1	1	0	1	1	0	1	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0



Logic Circuit for Binary to Gray Code Converter

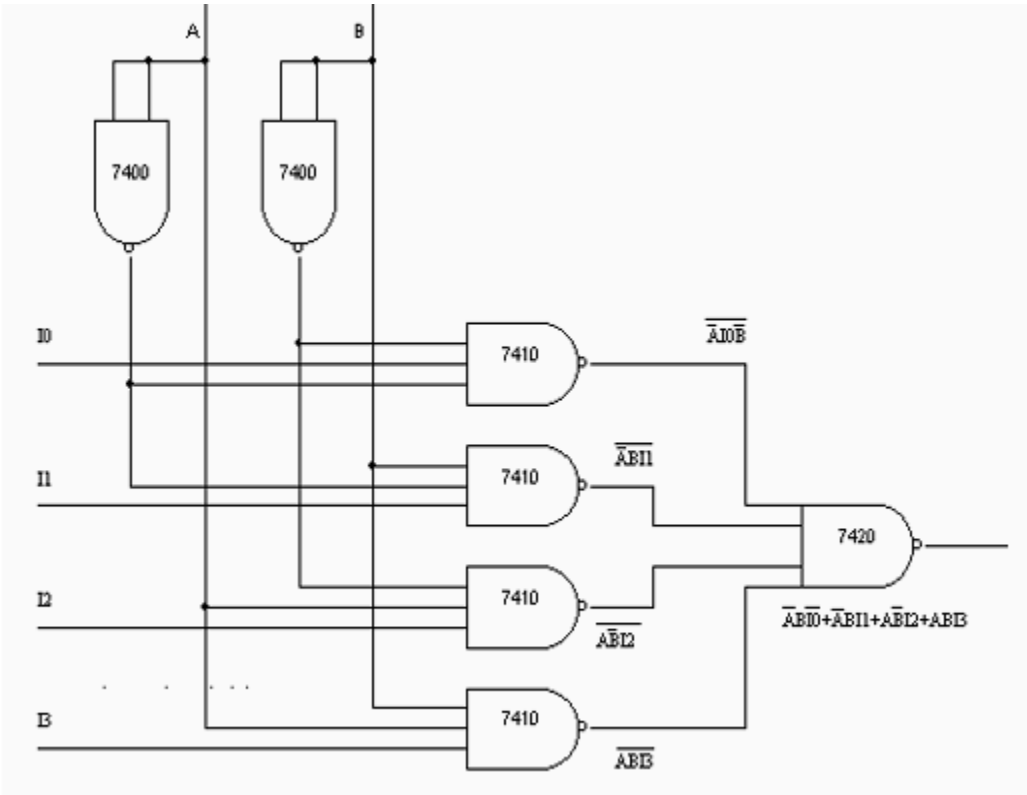


Logic Circuit for Gray to Binary Code Converter

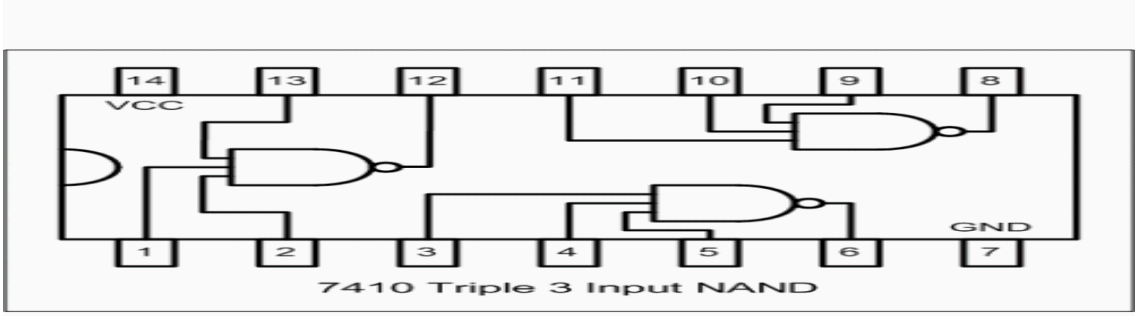
Results: The truth tables are verified.

EXPERIMENT NO.5

Design and implement
a. 4:1 Multiplexer (MUX) using only NAND gates.
b. Boolean expressions using 8:1 Multiplexer(MUX).

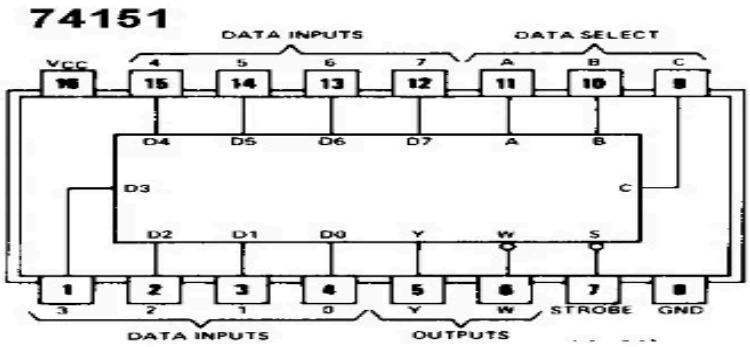


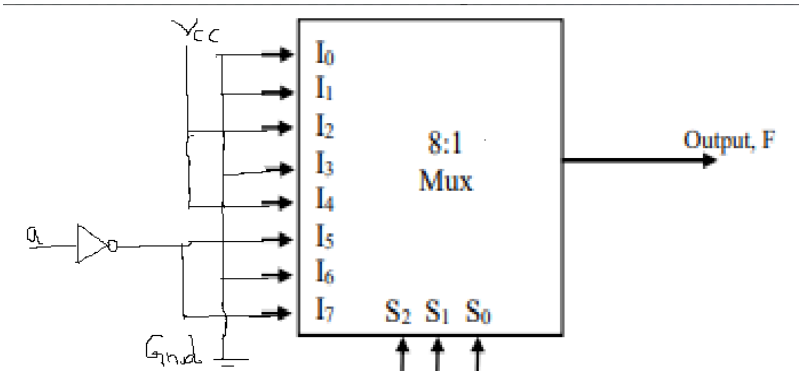
INPUT						OUPUT
A	B	I0	I1	I2	I3	Y (V)
0	0	0	X	X	X	0
0	0	1	X	X	X	1
0	1	X	0	X	X	0
0	1	X	1	X	X	1
1	0	X	X	0	X	0
1	0	X	X	1	X	1
1	1	X	X	X	0	0
1	1	X	X	X	1	1



Boolean expressions using 8:1 Multiplexer(MUX)

f(a,b,c,d) = Σm(2,4,5,7,10,14) using 8:1 MUX.

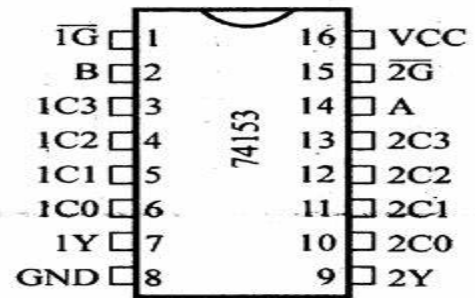




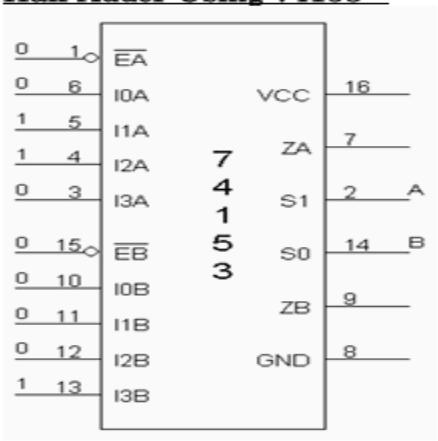
Results: The truth tables are verified.

EXPERIMENT NO.6

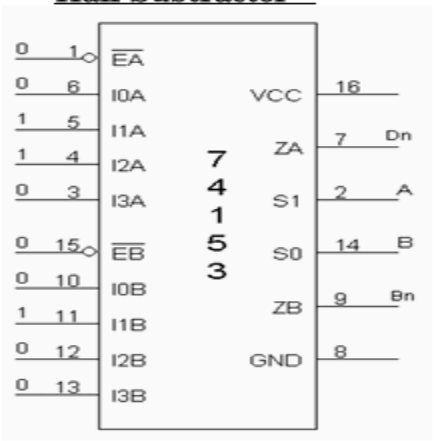
Half/Full Adder and Half/Full Subtractor using IC 74153.



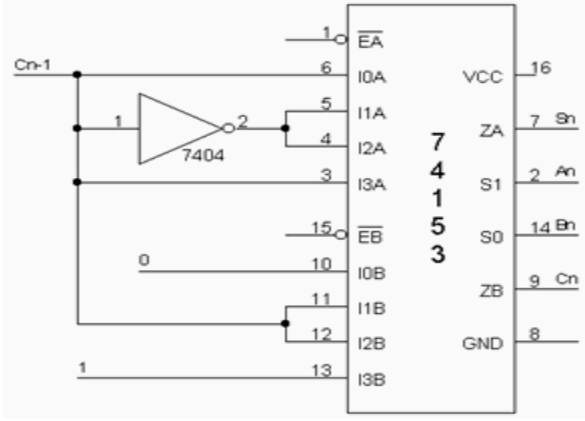
Half Adder Using 74153 –



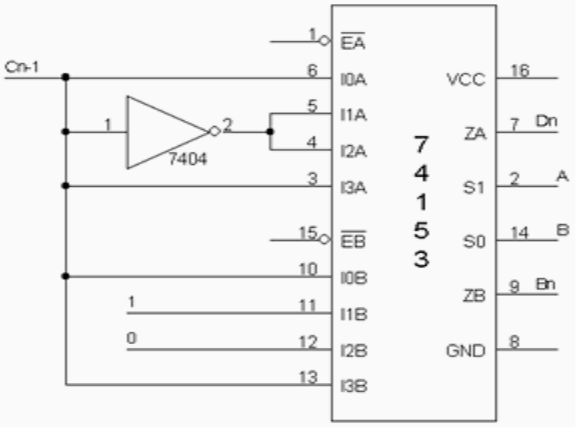
Half Subtractor: -



Full Adder Using 74153: -



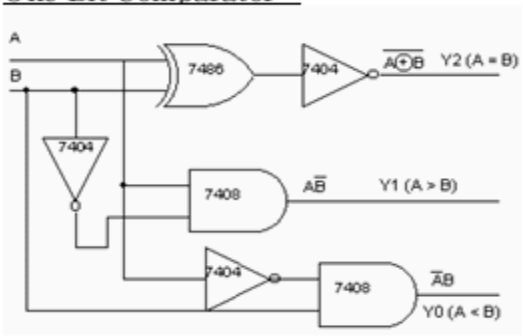
Full Subtractor Using 74153: -



EXPERIMENT NO.7

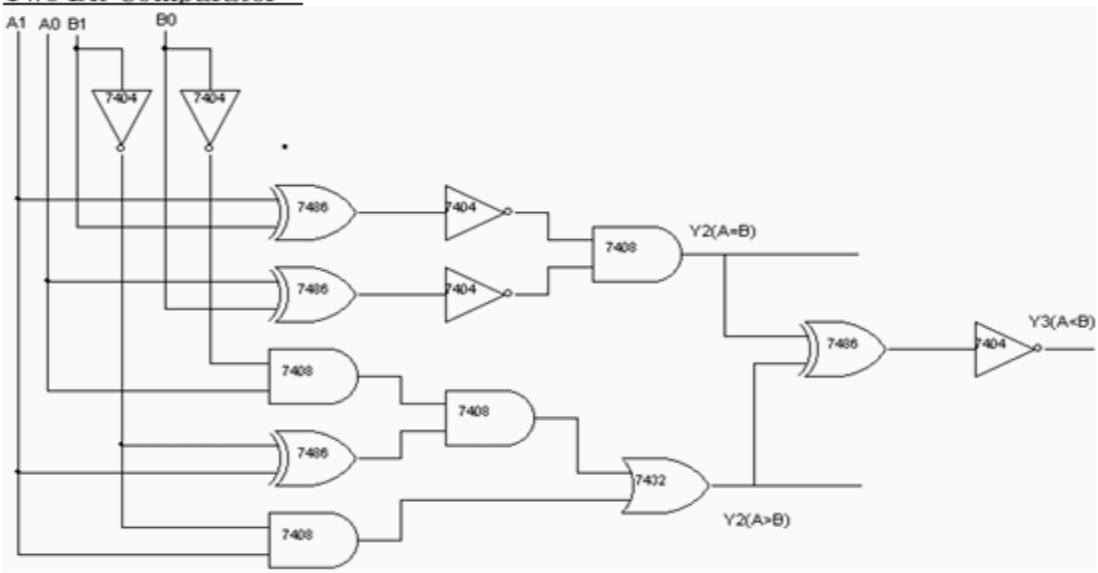
Design and implement One & Two-Bit Magnitude Comparator.

One Bit Comparator: -



A	B	Y1 (A>B)	Y2 (A = B)	Y3 (A < B)
0	0	0	1	0
0	1	0	0	1
1	0	1	0	0
1	1	0	1	0

Two Bit Comparator: -



Two-Bit Comparator: -

A1	A0	B1	B0	Y1 (A > B)	Y2 (A = B)	Y3 (A < B)
0	0	0	0	0	1	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	1	0	0
0	1	0	1	0	1	0
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	0	1	0
1	0	1	1	0	0	1
1	1	0	0	1	0	0
1	1	0	1	1	0	0
1	1	1	0	1	0	0

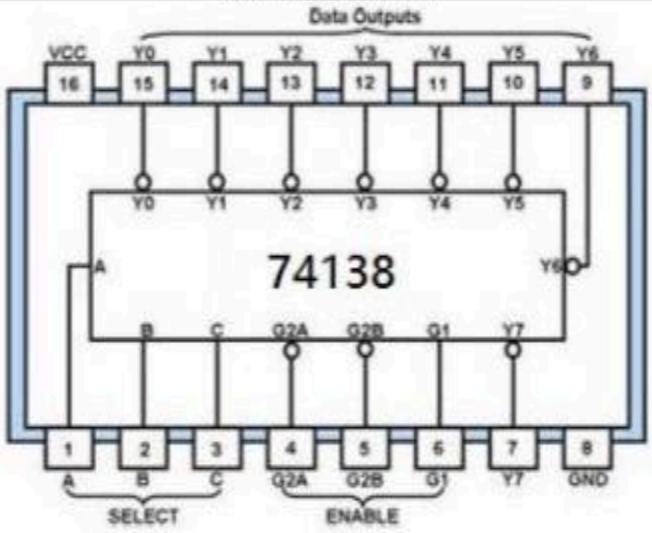
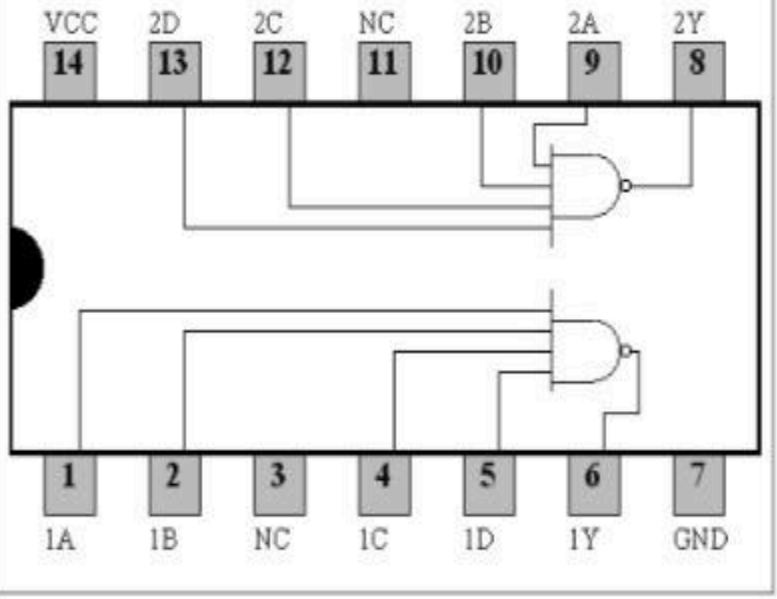
Results: The truth tables are verified.

EXPERIMENT NO.8

Design and implement

a. Decoder circuit using basic gates and to verify using IC74LS138.

b. Half/Full Adder and Half/Full Subtractor using IC74LS138.

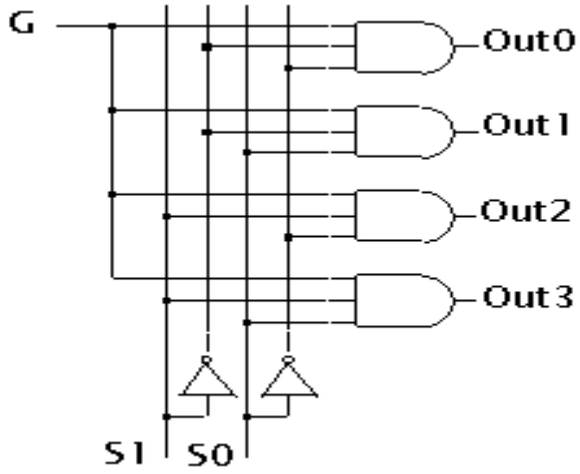
IC Name	Data Sheet Diagram
74138 **3-to-8 Decoder**	
7420 – 4input NAND	

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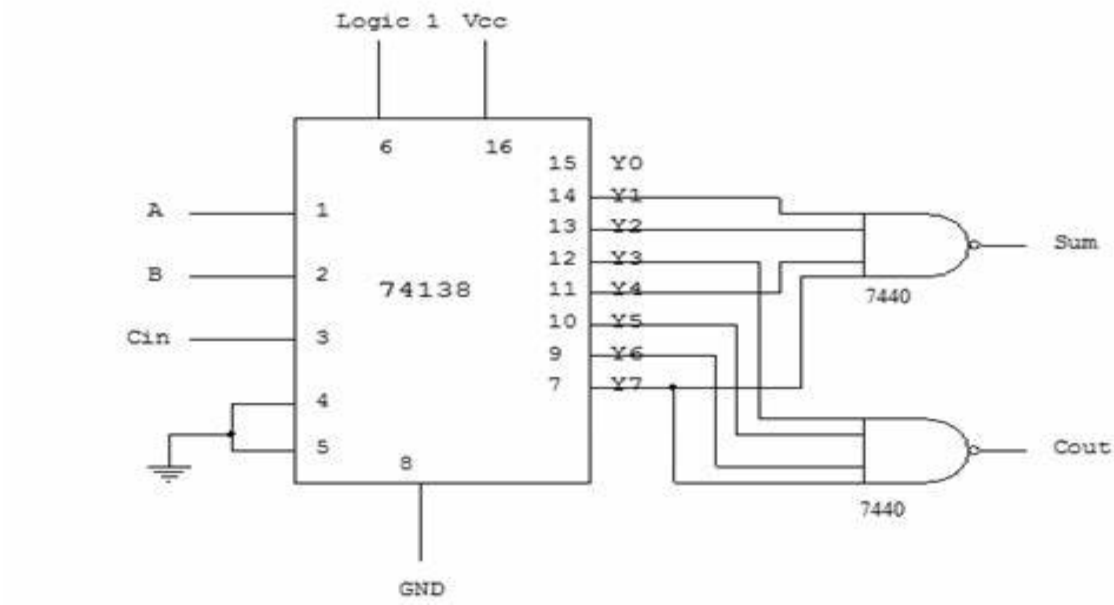
INPUTS					OUTPUTS							
ENABLE		SELECT										
G1	G2A,G2B	C	B	A	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7
H	L	L	L	L	L	H	H	H	H	H	H	H
H	L	L	L	H	H	L	H	H	H	H	H	H
H	L	L	H	L	H	H	L	H	H	H	H	H
H	L	L	H	H	H	H	H	L	H	H	H	H
H	L	H	L	L	H	H	H	H	L	H	H	H
H	L	H	L	H	H	H	H	H	H	L	H	H
H	L	H	H	L	H	H	H	H	H	H	L	H
H	L	H	H	H	H	H	H	H	H	H	H	L
X	H	X	X	X	H	H	H	H	H	H	H	H
L	X	X	X	X	H	H	H	H	H	H	H	H

H - HIGH, L - LOW, X - DONT CARE

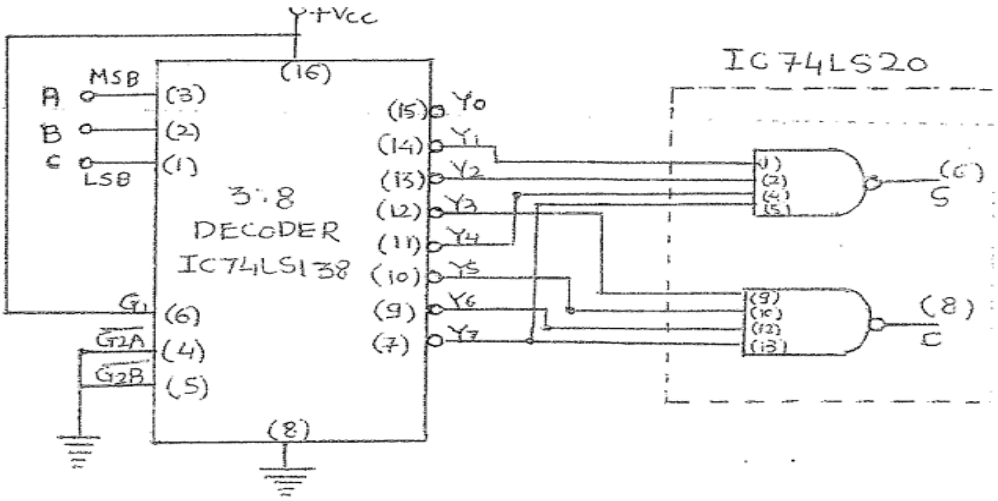
Decoder using logic gates



Circuit Diagram:



Full Adder

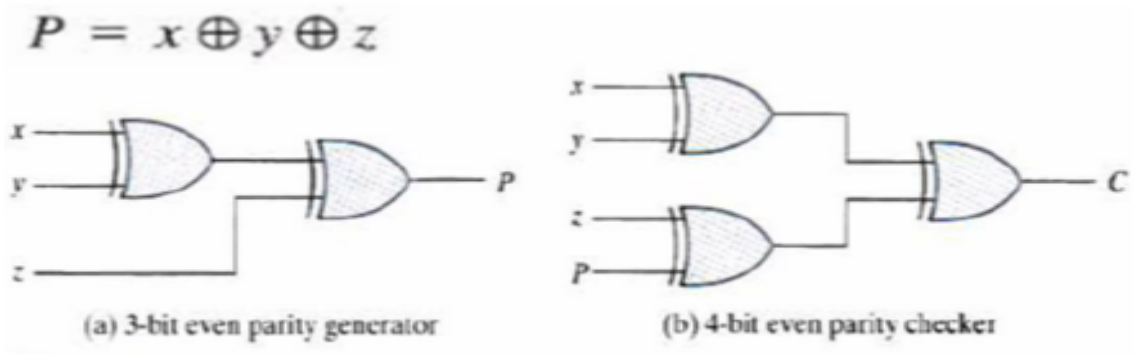


Full Subtractor

Results: The truth tables are verified.

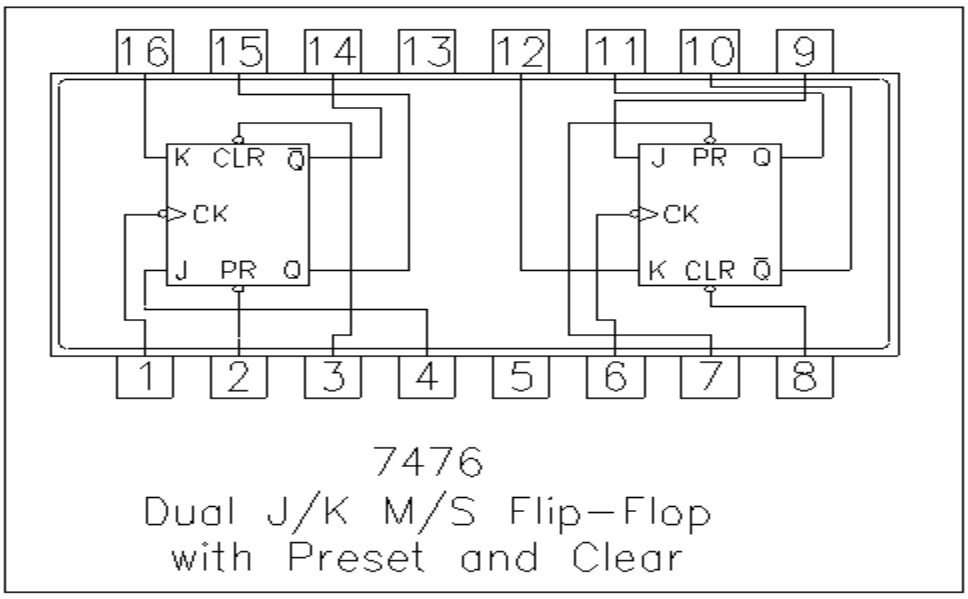
EXPERIMENT NO.9

a. Design and implement a 3-bit parity generator and verify using Parity Checker.
b. Implementation and verification of truth table for J-K flip-flop, Master-slave J- K flip-flop, D flip-flop and T flip-flop.





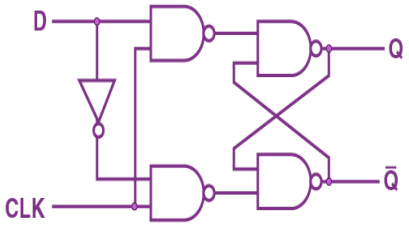
Implementation and verification of truth table for J-K flip-flop, Master-slave J- K flip-flop, D flip-flop and T flip-flop.



FUNCTION TABLE

INPUTS					OUTPUTS	
$\overline{PRE}$	$\overline{CLR}$	CLK	J	K	Q	$\overline{Q}$
L	H	X	X	X	H	L
H	L	X	X	X	L	H
L	L	X	X	X	H ↑	H ↑
H	H	⎍	L	L	Q_0	$\overline{Q}_0$
H	H	⎍	H	L	H	L
H	H	⎍	L	H	L	H
H	H	⎍	H	H	TOGGLE	

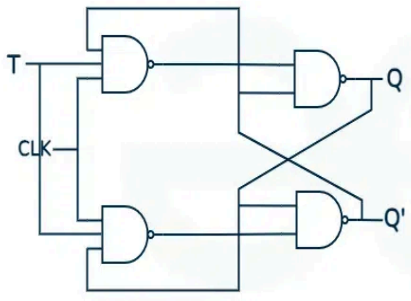
BYJU'S The Learning App



Truth Table

Q	D	$Q_{(t+1)}$
0	0	0
0	1	1
1	0	0
1	1	1

T flip flop



Truth Table

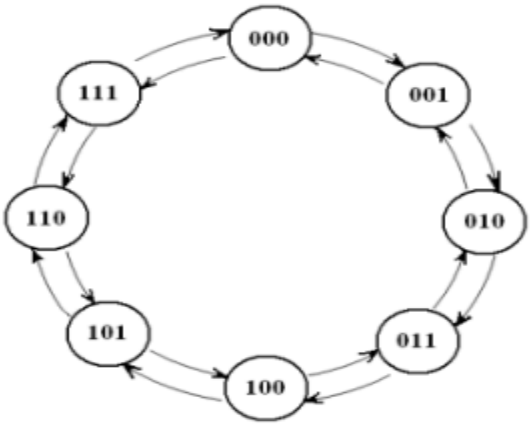
Q	Q_N	$Q_{(N+1)}$
0	0	0
0	1	1
1	0	1
1	1	0

Application of flip flop

Results: The truth tables are verified.

EXPERIMENT NO.10

Design and implementation of Mod-N synchronous counter using J-K flip-flops.



TRUTH TABLE:

Input	Present State			Next State			A		B		C	
Up/Down'	Q_A	Q_B	Q_C	Q_{A+1}	Q_{B+1}	Q_{C+1}	J_A	K_A	J_B	K_B	J_C	K_C
0	0	0	0	1	1	1	1	x	1	x	1	x
0	1	1	1	1	1	0	x	0	x	0	x	1
0	1	1	0	1	0	1	x	0	x	1	1	x
0	1	0	1	1	0	0	x	0	0	x	x	1
0	1	0	0	0	1	1	x	1	1	x	1	x
0	0	1	1	0	1	0	0	x	x	0	x	1
0	0	1	0	0	0	1	0	x	x	1	1	x
0	0	0	1	0	0	0	0	x	0	x	x	1
1	0	0	0	0	0	1	0	x	0	x	1	x
1	0	0	1	0	1	0	0	x	1	x	x	1
1	0	1	0	0	1	1	0	x	x	0	1	x
1	0	1	1	1	0	0	1	x	x	1	x	1
1	1	0	0	1	0	1	x	0	0	x	1	x
1	1	0	1	1	1	0	x	0	1	x	x	1
1	1	1	0	1	1	1	x	0	x	0	1	x
1	1	1	1	0	0	0	x	1	x	1	x	1

EXCITATION TABLE: (JK Flip-Flop)

Q	Q_{t+1}	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

K-MAP SIMPLIFICATION:

For J_A

UD QA \ QB QC	00	01	11	10
00	1	0	0	0
01	X	X	X	X
11	X	X	X	X
10	0	0	1	0

$J_A = \overline{UD}\ \overline{QB}\ \overline{QC} + UD\ QB\ QC$

For J_B

UD QA \ QB QC	00	01	11	10
00	1	0	X	X
01	1	0	X	X
11	0	1	X	X
10	0	1	X	X

$J_B = \overline{(UD \oplus QC)}$

For K_A

UD QA \ QB QC	00	01	11	10
00	X	X	X	X
01	1	0	0	0
11	0	0	1	0
10	X	X	X	X

$K_A = \overline{UD}\ \overline{QB}\ \overline{QC} + UD\ QB\ QC$

For K_B

UD QA \ QB QC	00	01	11	10
00	X	X	0	1
01	X	X	0	1
11	X	X	1	0
10	X	X	1	0

$K_B = \overline{(UD \oplus QC)}$

For J_C

UD QA \ QB QC	00	01	11	10
00	1	X	X	1
01	1	X	X	1
11	1	X	X	1
10	1	X	X	1

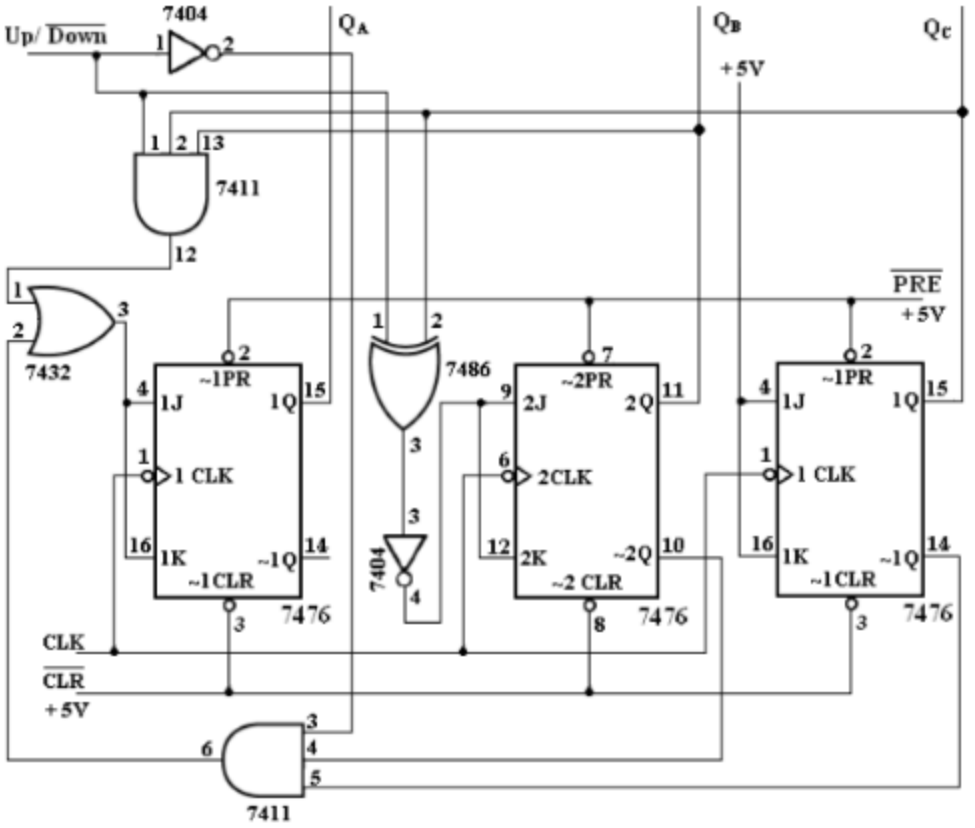
$J_C = 1$

For K_C

UD QA \ QB QC	00	01	11	10
00	X	1	1	X
01	X	1	1	X
11	X	1	1	X
10	X	1	1	X

$K_C = 1$

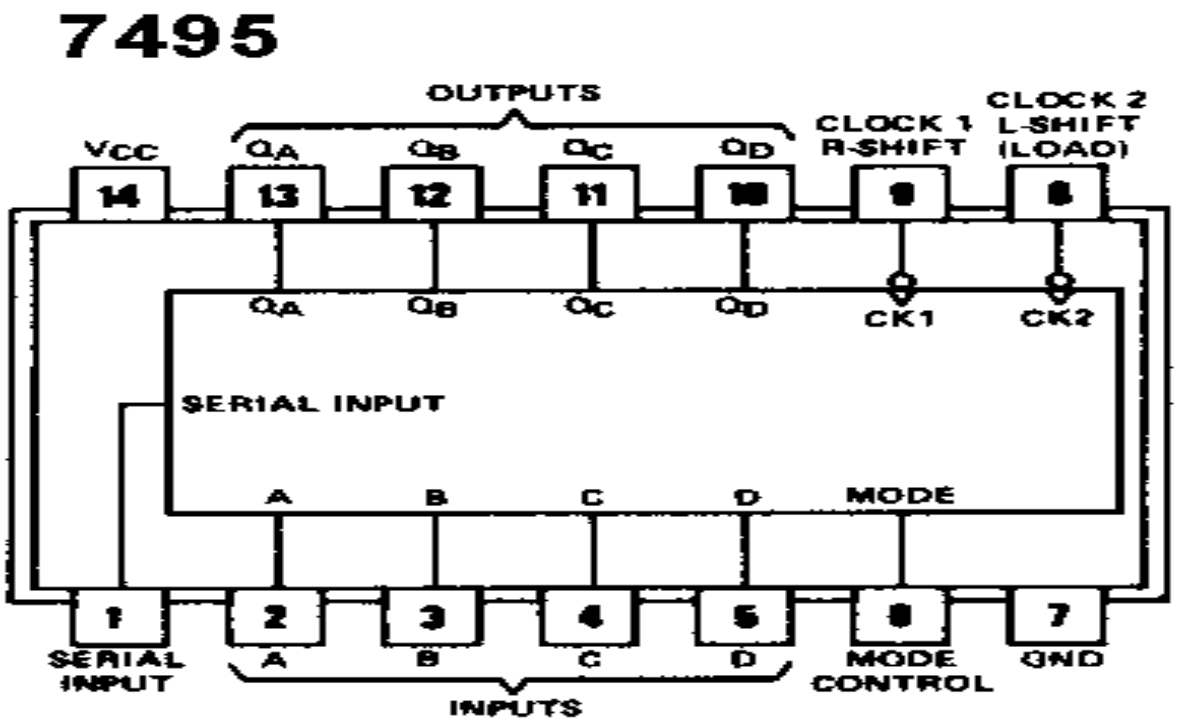
LOGIC DIAGRAM:



Results: The truth tables are verified.

EXPERIMENT NO.11

Design and implementation of shift register to function as
i) SISO ii) SIPO iii) PISO iv) PIPO v) shift left and vi) shift right operation.



PROCEDURE:
1. Derive the wiring diagram.
2. Connect Input terminals to the Logic Input Sockets.
3. S1,S0, SERIAL LEFT,SERIAL RIGHT,PARALLEL DATA-A,B,C and D to Logic Input Sockets.
4. Connect CLEAR to VCC.
5. Connect External Clock to the CLOCK Terminal.
6. Connect QA, QB, QC and QD to the Logic Output terminals.
7. Observe output changes at O/P Terminals with given Truth Table.

Mode Control		Register Type
S_1	S_0	
0	0	Hold
0	1	Shift to right
1	0	Shift to left
1	1	Parallel Mode

INPUTS										OUTPUTS			
CLEAR	MODE		CLOCK	SERIAL		PARALLEL				QA	QB	QC	QD
	S1	S0		LEFT	RIGHT	A	B	C	D				
H	X	X	⌐	X	X	X	X	X	X	QA0	QB0	QC0	QD0
H	H	H	⌐	X	X	a	b	c	d	a	b	c	d
H	L	H	⌐	X	H	X	X	X	X	H	QAn	QBn	QCn
H	L	H	⌐	X	L	X	X	X	X	L	QAn	QBn	QCn
H	H	L	⌐	H	X	X	X	X	X	QBn	QCn	QDn	H
H	H	L	⌐	L	X	X	X	X	X	QBn	QCn	QDn	L
H	L	L	X	X	X	X	X	X	X	QA0	QB0	QC0	QD0

CONCLUSION: Verified the Truth Table of Universal Shift Register.

EXPERIMENT NO.12

a. Design and implementation of i) Ring counter and ii) Johnson counter using 4-bit shift register.

b. Design and implementation of Decade Counters

c. Design and implementation of Asynchronous Counters

Ring Counter Truth Table

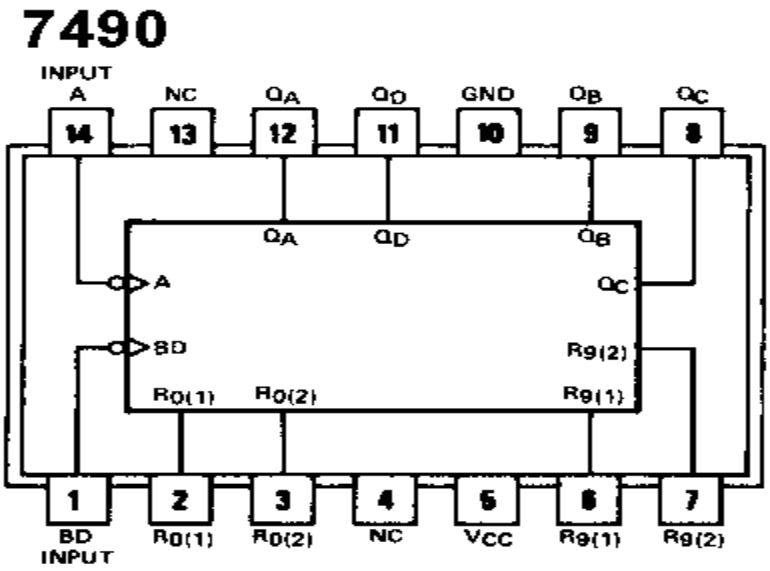
PRESETED 1

ORI	CLK	Q0	Q1	Q2	Q3
low	X	1	0	0	0
high	low	0	1	0	0
high	low	0	0	1	0
high	low	0	0	0	1
high	low	1	0	0	0

Johnson Counter Truth Table

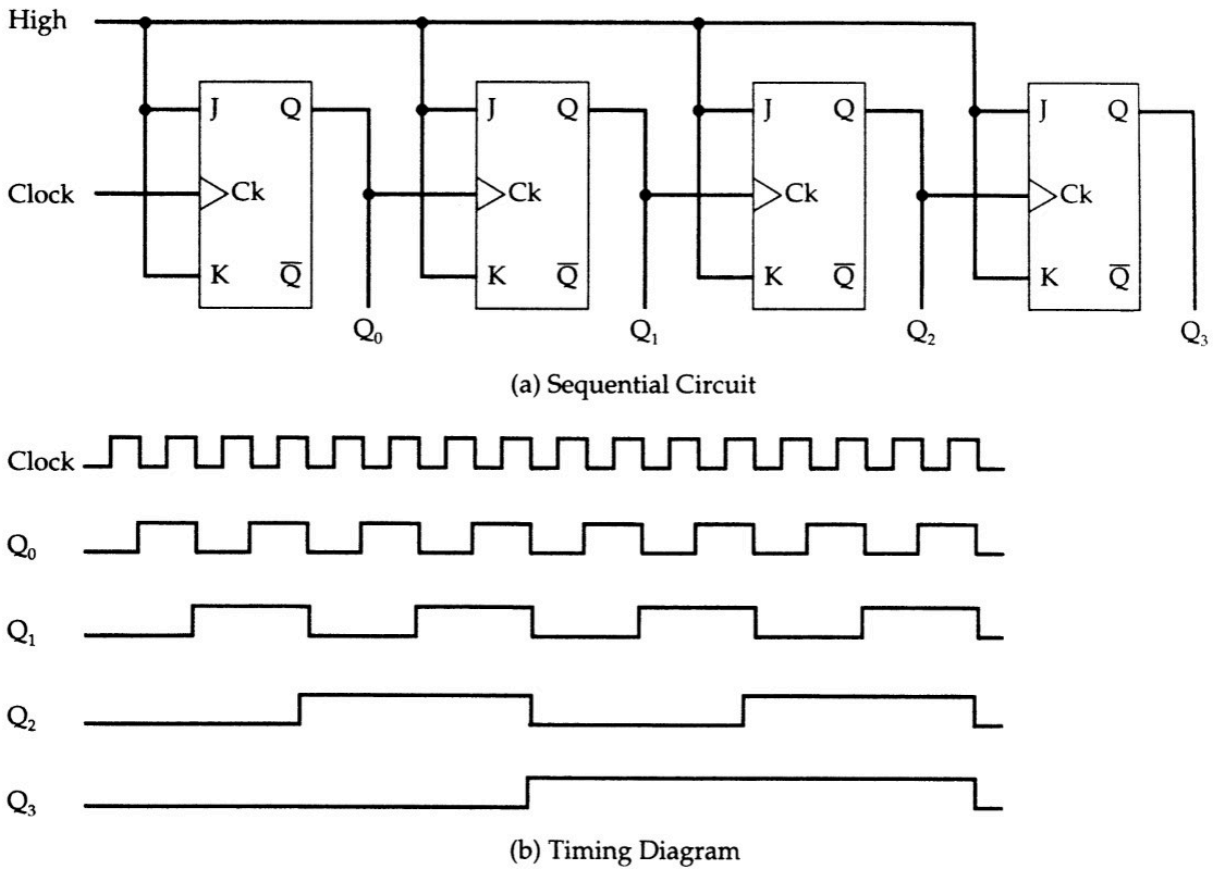
Clock Pulse	Q_0	Q_1	Q_2	Q_3
0	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1
5	0	1	1	1
6	0	0	1	1
7	0	0	0	1

Decade Counter



Count	Qd	Qc	Qb	Qa
(Start) 0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
(New Cycle) 10	0	0	0	0

Asynchronous Counter



Results: The truth tables are verified.